Assignment- 4

Note: You are advised to use modular programming approach with consideration of basic functionalities of JAVA (not build up libraries).

1. Design a utility based program that maintain your record containing Name, Father's Name, Date of Birth, and Age (computed with respect to the current date and displayed in years, months, and days), registration number, qualifying degree with % marks where registration number can be in string or integer or long type. Accept input using the Scanner class and display each field on a new line with appropriate labels. Also, used the concept of immutable object for ***AgeUtility*** that accepts date of birth and compute the age.

2. Repeat the problem No 1 utilizing the concept of generic class.

3. Design a utility based reusability concept that computes the perimeters, areas of rectangle, circle, rhombus where input parameters are accepted as int, float, long, double or its combinations. Further, you are required to use constructor overloading for providing flexibilities in object initializations. Use concept of immutable class that preserve the value of input parameters. Display the computed results.

4. Repeat the problem No 3 utilizing the concept of generic class.

5. Design a immutable class based program that compute the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) for five integers. The integer numbers may be initialized using blocked constructor or through run time inputs. The once initialized input parameters its value to be required to be preserved (un-modified). Your program may have all possible ways the initialized values are preserved. Display the computed results along with concerned input parameters.